

A Data-Driven Variance Reduction Technique for Efficiently Modelling Astronaut Radiation Doses in Spacecraft in High-Energy Isotropic Radiation Fields

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The ionizing radiation exposure to crew on current and future space missions can significantly increase their health risks for cancers, degenerative diseases, and other acute and late effects. A common approach for estimating risk to crew is by completing stochastic (e.g., Monte Carlo) or deterministic particle transport simulations. Within the simulated environment, a small fraction of the particle histories tracked will interact with the astronaut or detector, particularly for larger spacecraft such as the International Space Station, Tiangong Space Station or Lunar Gateway. These simulations can be computationally intensive as they require a very large number of particle histories to achieve a low statistical uncertainty. Variance reduction techniques are applied to simulations to reduce the computational time of the simulation while maintaining the same (or less) statistical uncertainty. The variance reduction technique developed herein involves applying a directional source bias to an isotropic radiation field, such as galactic cosmic rays, to reduce the quantity of particles that have a low probability of interacting with the astronaut or detector. A custom application has been developed utilizing the Geant4 Toolkit that computes the trajectories and energies of particles in three dimensions in the International Space Station using the Monte Carlo method. The results demonstrate the impact of our variance reduction technique on effective dose equivalence depending on: primary and secondary particle type (proton, neutron, photon, heavy ion, etc.), geometric volumes and spacecraft materials. Our variance reduction technique can be tuned by the user to optimize the simulation time depending on their objectives and enables rapid testing of different shield configurations and materials. This variance reduction technique is implemented easily using several input parameters for boundary conditions. Recommended values are presented for rapid implementation in simulations. © 2023 by Radiation Research Society

INTRODUCTION

Since the year 2000, humanity has maintained a constant human presence in space. This time was spent in low Earth orbit (<450 km altitude) where there is significant protection from solar ionizing radiation by the Earth's geomagnetic field. The first astronauts wore personal dosimeters until it was discovered that their radiation exposure after a 6 month stay at the International Space Station (ISS) was not significantly higher than the dose received by a commercial airline pilot over their career (1). Today, dosimeters are attached to the inside of habitable modules on the ISS and are monitored yearly (2). Nevertheless, radiation remains the primary health concern for astronauts in space (3) as radiation health effects can range from early signs of radiation sickness to cancer induction (4). Other possible effects include central nervous system damage (5), cataracts (6), cardiovascular damage (7), impaired wound healing (8), and infertility. Extensive effort has been put into developing materials to attenuate dose on astronauts due to space radiation with novel approaches being applied to both passive (9) and active shielding technologies (10). It remains a particular challenge to construct a lightweight habitat (to reduce launch masses) which simultaneously protects crew against radiation, meteoroid impacts and extreme fluctuations in temperature while remaining pressurized. The currently accepted solution utilizes multiple materials, in a layered configuration, which inevitably renders simulations complex and time consuming.

The ionizing radiation field in space is highly complex due to the multitude of particles over an extreme energy range, temporal variations, and the environmental dynamics. Galactic cosmic rays (GCR), solar energetic particles (SEP) and trapped protons and electrons in the Van Allen Belts all contribute to the ionizing radiation field.

GCR consist of high-energy protons and atomic nuclei and are an abundant source of ionizing radiation outside of Earth's magnetic field. The baryon component is composed of approximately 87% protons, 12% helium ions (alpha particles) and the remaining 1% are heavy ions ($Z > 2$) (11). The Sun's magnetic field extends throughout the

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interplanetary medium, changing the velocity of GCR as they traverse through it. Throughout the 11-year solar cycle, the magnetic field orientation of the Sun changes, resulting in a variation in GCR energies. The highest energies occur during a solar minimum period (12). SEP's are particles emitted by the sun, accelerated by either solar flares or coronal mass ejections (CME) (13). Similar to GCR, CME include protons and heavy ions from the keV to the GeV energy range, however they can have a significantly higher flux. It is well understood that there is a large contribution to radiation dose from secondary particles, most notably; neutrons, photons (gamma rays), protons, electrons and pions. Fragmentation is a common nuclear interaction that produces secondary particles from impacts of nucleus-nucleus collisions by high energy incident particles. Bremsstrahlung is a dominant process for photon production by charged particles and can produce secondary photons isotropically. Due to the difficulty of replicating the space radiation environment for Earth-based experiments, the most common approach in research is to use particle transport simulation software (14). Two methods are most commonly used. The first is the Monte Carlo method on which several applications are based [Geant4, FLUKA (15), Monte Carlo NP]. The second option involves solving the Boltzmann Transport equations, and is most commonly implemented using NASA's High charge (Z) and Energy TRAnsport (HZETRN) deterministic transport code (16). Geant4 Monte Carlo simulations of the radiation environment on board the ISS have been completed to estimate radiation dose to astronauts (17). These simulations can be computationally intensive as they require a very large number of particle histories to achieve a low statistical uncertainty in the quantities of interest. Within the simulated environment, only a small fraction of the tracked particles will interact with the detector, particularly for larger spacecraft such as the ISS, Tiangong Space Station or Lunar Gateway. The computational time is dependent on several factors including the number of particles (or histories) modelled, the particle type (proton, neutron, pion, heavy ion, etc.), quantity of geometric volumes, detector size and the spacecraft materials (18). It is preferable to implement techniques to reduce computation time without unduly compromising accuracy.

Variance reduction techniques (VRT) involve reducing the computational time of a simulation while maintaining the same (or lower) statistical uncertainty. At the time of writing, VRTs have not been studied for Monte Carlo simulations in the space radiation environment. This work presents a novel procedure for applying a directional source bias to an isotropic radiation field, such as GCR particles. The approach is analogous to those employing directional biasing and range rejection in Monte Carlo codes applied to medical radiation exposures (19, 20).

A custom application was developed for this study utilizing Geant4. Geant4 is a toolkit developed by CERN that computes the trajectories, interactions and energetics of particles in three dimensions using the Monte Carlo method (21). A wealth of

literature exists for the unit testing and benchmarking of each cross-section or model to experimental data (22, 23).

When a spacecraft is within the GCR radiation environment, secondary particles are produced in all materials through numerous interactions and processes. Extremely high-energy particles traverse the hull, stowage, payload and other internal materials, undergoing particle interactions producing a shower of secondary particles and chain reactions. As a result, a complex secondary environment is produced consisting of protons, neutrons, photons, pions, muons, ions and more. In the simulated environment, if the modelled isotropic radiation field is large enough to encompass the entire spacecraft, the contributions from all possible interactions and processes can be computed if the simulation is run for long enough. However, the full simulation is relatively inefficient, the further the material is from the detector, the lower the probability that an incident particle or any subsequent particle produced will reach the detector. For example, there is an extremely low probability of a secondary particle generated in the Zvezda Service Module or solar array hitting a detector or astronaut in the Columbus module. When simulating all possible outcomes, a significant portion of computation time is spent tracking particles that will not contribute to reported dose. To overcome this, the isotropic radiation field does not need to encapsulate the entire spacecraft, but rather only needs to be sufficient in size to simulate particle interactions and secondaries that will be major contributors to dose. By constraining the radius of influence, computation time can be reduced substantially while not compromising the calculated dose beyond the uncertainties normally accepted in radiation protection contexts. The objective of the present work is to describe a method for simulating an isotropic radiation field around a location of interest of a spacecraft, quantify dose contribution for primary and secondary particles and to calculate the optimal isotropic region radius for a given error tolerance. The efficiency of the simulations is compared to full (analog) Monte Carlo simulations to quantify the effectiveness of the implemented VRT.

METHODS

A full (analog) Monte Carlo simulation of radiation transport through the ISS has been conducted with detectors placed in four different modules including Columbus, Node 1, Destiny and the Zvezda Service Module.

For each particle that contributes dose in a detector, the particle species, energy, and the first interaction location of the primary particle was recorded. Cumulative distribution plots are generated for the percentage contribution of effective dose equivalent as a function of the first interaction point radius (denoted by R_1) for each location, secondary particle type and primary particle species. For VRT implementation, an isotropic region radius, R_{ISO} , smaller than that in the analog simulation can be used. To quantify the effectiveness of the VRT, an efficiency measure is adopted and has been calculated for 10 isotropic region radii. The following section discusses in detail the geometry and materials, physics, boundary conditions and scoring of the Monte Carlo simulations.

Geometry and Materials

Early simulations conducted by the authors generated dose distributions within habitable modules on the ISS due to radiation impacts

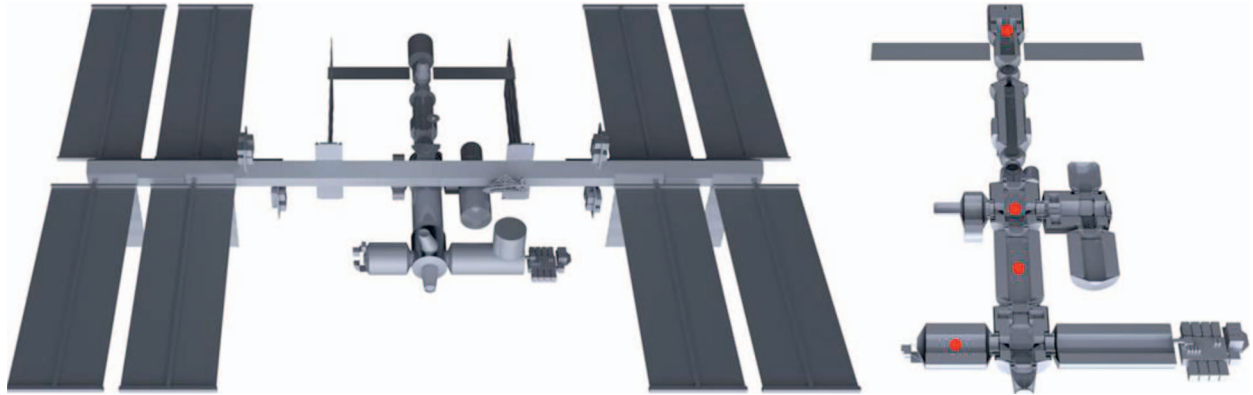


FIG. 1. CAD model of International Space Station (2020 configuration) developed in SolidworksTM for use in the custom Geant4 simulations. The right image is a cross-section of the model showing the interior. The four detectors used in the simulations are indicated by the red dots in (from top to bottom) Zvezda Service Module, Node 1 (Unity), Destiny and Columbus.

(24). The resulting inhomogeneous dose distributions identified potential locations for the placement of additional radiation shielding that would provide the largest dose reduction for astronauts. The model of the ISS used, however, was low fidelity and only included a limited number of materials. Here, a 1:1 model of the 2020 configuration of the ISS has been created using CAD software and utilized for simulations as it resembles the typical size, shape and materials of current and future crewed spacecraft. The geometry was developed in SolidworksTM and contains a total of 199 parts. Each module contains 5 to 10 solids consisting of the exterior shell of aluminum, an intermediate layer of Kevlar and epoxy, and an inner shell of aluminum. The layering of the spacecraft was designed to resemble the design proposed by Christiansen et al. (25). There is very little information publicly available on structures and materials of interior components. Interior hardware can include payloads, payload racks, video cameras, data and mission computers, fire extinguishers, thermal control systems and valves, coolant and pipes, ventilation, power distribution units, heat exchangers, sleeping quarters and more. It is not possible to model these components and calculate their impact on astronaut dose. The selected approach was to approximate complex structures as homogeneous volumes of an appropriately chosen elemental mixture with an assigned density that reflects the total mass. The selected materials include aluminum, Kevlar[®], water, copper and plastic. It is a technique previously implemented successfully in modelling radiation and scattering shielding properties of complex systems such as a medical linear accelerator and the ISS (17, 26). The ISS model is of moderate fidelity and is a reasonable approximation for general radiation studies. Increasing the geometry fidelity further can result in larger computation time with minimal impact on dose calculations (17). While the referenced study was completed for simulations in the Columbus module, it is assumed to be true for other modules. Figure 1 is a render of the complete geometry before it was imported into the custom Geant4 application. The density of the interior solids was modified to reproduce the total module mass within 1% of launch values (27). Air was not included in the simulations and was assumed to have a negligible impact on the radiation environment.

Physics

Geant4 has a number of predefined physics lists that specify the hadronic elastic and inelastic models, capture, fission processes, cross sections and interactions that are included in the simulations. The physics list recommended for space applications is “QBBC” and was used for all simulations (22, 28, 29). The commonly invoked hadronic inelastic models include Bertini Intranuclear Cascade (BERT), Binary Light Ion Cascade (BIC), Fritiof model (FTF) and Liege Intranuclear Cascade model (INCL++) (30, 31). Inelastic nucleus-nucleus scattering is invoked by the BIC (0–4 GeV/n) and FTF (2 GeV/n–100 TeV/n) models (32). Validation against experimental data is presented in the Geant4 fragmentation models for

Heavy Ion Therapy (23). Full details of all physics components i.e., hadronic, electromagnetic, decay etc, included in the reference physics lists are provided on the GEANT website (32). The range cut for protons, pions, photons and electrons used throughout the experiments was 0.1 mm.

Empirical models of nuclear reactions (including fragmentation and spallation) for a wide variety of particles and energies have been developed and validated using particle accelerators (22, 23).

Boundary Conditions

The GCR spectrum (solar minimum 2010) has been generated by the BON2020 model and has been used for the radiation input for all the simulations (33). This model defines the energy distribution and relative abundances of GCR particle species from hydrogen to nickel. Figure 3 is a plot of the GCR spectrum for hydrogen, helium, oxygen and iron ions. The maximum particle energy modelled for all incident GCR species is 50 GeV/n and therefore the inelastic nucleus-nucleus scattering models can be invoked for all particles. All particles simulated are assumed to be fully ionized.

The particles are generated on the surface of the “world sphere” with radius R_W . The angular distribution of particles follows a cosine distribution to ensure the fluence is isotropic and is constrained to within an angle (Θ_{max}). Θ_{max} represents the maximum incident angle from the position the particle is generated (P_G) and the center of the detector (P_D). The isotropic region refers to a sphere in which the particles are approximately isotropic with a radius R_{iso} . For the analog simulation of the ISS, the region must be large enough to encapsulate the entire spacecraft, denoted by $R_{iso} = R_{isomax} = 100$ m.

These parameters are illustrated in Fig. 2. Θ_{max} is an input parameter in Geant4’s general particle source that was used to define the isotropic radiation field. These parameters are related by a simple sine rule as given in Eq. (1).

$$\Theta_{max} = \text{asin}\left(\frac{R_{isomax}}{R_W}\right) \quad (1)$$

The requirements for generating a reasonable approximation of an isotropic radiation environment include applying a cosine angular distribution and using a R_W to R_{iso} ratio at least 100. At all positions within the isotropic sphere, the maximum variation in flux is 1%. More details of the mathematics used are included in the Supplementary Materials² (<https://doi.org/10.1667/RADE-23-00027.1.S1>).

For the full analog simulation results presented, an R_W of 10 km and R_{isomax} of 100 m is used. This R_{isomax} was selected to encapsulate the entire ISS in the isotropic radiation field. A bias was applied to

² Editor’s note. The online version of this article (DOI: <https://doi.org/10.1667/RADE-23-00027.1>) contains supplementary information that is available to all authorized users.

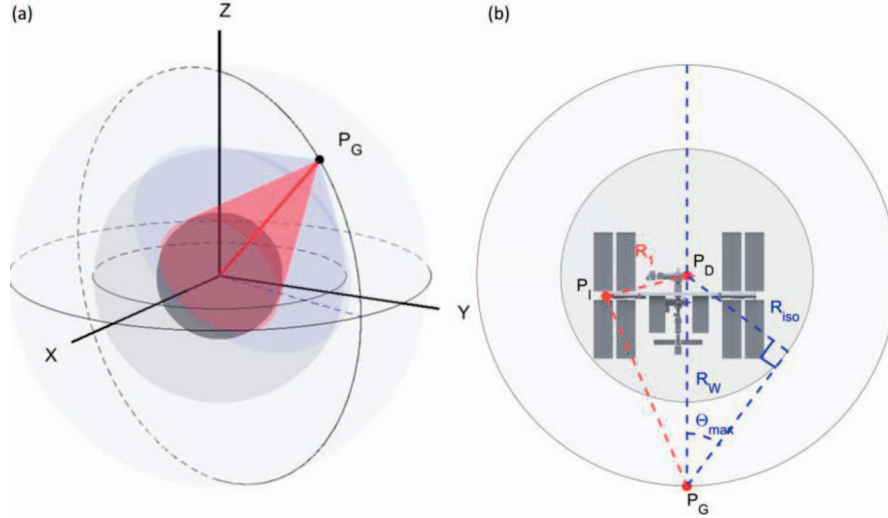


FIG. 2. Diagrammatic representation of the isotropic radiation field. Panel (a): P_G is an example starting position of a particle and the red cone represents all the possible angles its initial direction can take; panel (b): 2D representation of the isotropic region radius. The blue right-angled triangle shows the analog simulation parameters where $R_{iso} = R_{isomax}$. (R_W = world sphere radius; Θ_{max} = maximum angle between particle position, P_G and center of detector, P_D .) The red triangle compares the locations of the position of the primary particle's first interaction, P_I and the center of the detector, P_D .)

heavy ions to simulate higher than usual GCR abundances to reduce the statistical uncertainty in results.

The Supplementary Materials (<https://doi.org/10.1667/RADE-23-00027.1.S1>) also contains further mathematical equations that can be used to compute the vectors for the particle boundary conditions (position and direction) for the isotropic radiation source and variance reduction technique.

Scoring

Astronaut dose is estimated by scoring the fluence of particles reaching a spherical detector of radius 0.5 m after transport through the spacecraft and converting to dose using ICRP conversion factors. Detectors were placed in four locations including Columbus, Destiny, Node 1 and the Zvezda Service Module as shown in the right panel in Fig. 1.

To calculate effective dose equivalence, the isotropic (ISO) fluence-to-dose conversion coefficients published in ICRP-116 and ICRP-123 (34–36) and the NASA quality factors have been used. A quality factor (Q) reflects the relative biological effectiveness of high-linear energy transfer (LET) compared to low-LET radiation at low exposure levels. A constant quality factor of 1 was used for electrons, gamma rays and muons due to their low-LET. In the studies referenced in ICRP-123, tomographic data was used to produce a voxelized computational model representing a reference male and female (37). The PHITS Monte Carlo code was used to estimate the absorbed dose to each organ. The dose for each organ is a complex function of incident particle and energy. Fluence-to-absorbed-dose conversion coefficients for deuterons and tritons were obtained from (38).

The mean absorbed dose in organ, T for a radiation type, R can be obtained using Eq. (2) where $d_{T,R}(E)$ is the fluence-to-absorbed-dose conversion coefficient and ϕ_R is the fluence of particles incident on the detector.

$$D_{T,R} = \int_0^\infty d_{T,R}(E) \frac{d\phi_R}{dE} dE \quad (2)$$

The dose equivalent, $H_{T,Q}$ in organ, T is calculated using Eq. (3), where $Q_{T,R}$ is the mean quality factor in organ, T and radiation type, R .

$$H_{T,Q} = \sum_R \int_E Q_{T,R}(E) d_{T,R}(E) \frac{d\phi_R}{dE} dE \quad (3)$$

The effective dose equivalent (H_E) is the sum of the tissue-weighted dose equivalent, W_T of all organs and tissues in the body, averaged for males and females, calculated using Eq. (4).

$$H_E = \sum_T W_T H_{T,Q} \quad (4)$$

W_T represents the tissue weighting factors defined in ICRP-103 (36). The method for calculating effective dose equivalent is discussed in detail in ICRP-123 (35) and has been adopted by many space agencies around the world for completing radiation assessments. Throughout this paper, the term *dose* will be used for “effective dose equivalent” to simplify explanations.

To determine the distance at which secondary radiation contributions are significant, the position of the primary particle's first interaction is recorded for each particle that deposits dose. In this way, the cumulative distribution of dose to the detector can be plotted as a function of the distance from the primary trajectory. These plots can be exploited to develop a VRT by defining the R_{iso} required to achieve a given dose tolerance.

Two locations are required for developing and assessing the VRT; the position of the primary particle's first interaction ($P_I(x_i, y_i, z_i)$) and the center of the detector ($P_D(x_d, y_d, z_d)$) as indicated by the red triangle in Fig. 2b. These positions were recorded for each particle reaching the detector. The distance between the first interaction location and the center of the detector can be calculated using Eq. (5) and adopted as the independent variable for the cumulative dose distributions, which were generated for each location, secondary particle type and primary particle species.

$$R_1 = \sqrt{(P_{ix} - P_{dx})^2 + (P_{iy} - P_{dy})^2 + (P_{iz} - P_{dz})^2} \quad (5)$$

Variance Reduction and Efficiency

The variance reduction technique involves simulating a spherical isotropic radiation field of radius R_{iso} , where $R_{iso} < R_{isomax}$. An example is presented in Fig. 2a by the dark shaded sphere. Cumulative plots of dose against first interaction distance (R_1) represent the upper bound on the error in simulations conducted with R_{iso} less than R_{isomax} , and could also be used to provide estimated correction factors for recovering the full analog simulated dose. For simulations of

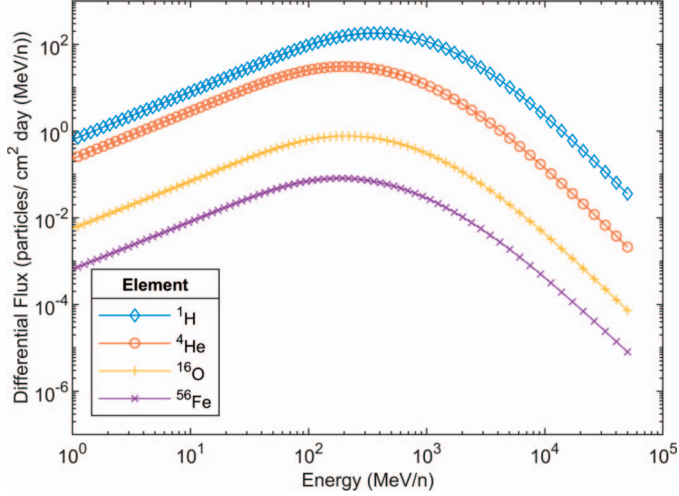


FIG. 3. Galactic Cosmic Ray Spectrum - BON2020 model - for the 2010 solar minimum (33). Data was generated and imported from OLTARIS (40).

the ISS or similar spacecraft geometry and materials, users can select an isotropic region radius from the results presented here, and implement the VRT directly using a maximum incident angle for each particle species calculated using Eq. (6).

$$\Theta_{max} = a \sin\left(\frac{R_{iso}}{R_W}\right) \quad (6)$$

To quantify the effectiveness of the VRT, a commonly used efficiency metric has been calculated. Efficiency (ε) is defined using Eq. (7).

$$\varepsilon = \frac{1}{\sigma^2 T} \quad (7)$$

where σ^2 is the variance in the quantity of interest and T is the computation time. Several simulations were completed with different isotropic radii and the computation time of each batch was recorded. The number of histories was scaled according to Eq. (8) for the solid angle, Ω , of the isotropic region to normalize for particle fluence.

$$\Omega = 4\pi \sin^2\left(\frac{\Theta_{max}}{2}\right) \quad (8)$$

Simulations at all angles retain 90% or more of total dose to the detector, and therefore the statistical variance in the recorded particles can be approximated as a constant, and the efficiency gains are described by $1/T$ according to Eq. (7). The GPS input commands for VRT implementation of GCR protons and alpha particles in Geant4 can be found in the Supplementary Materials (<https://doi.org/10.1667/RADE-23-00027.1.S1>).

RESULTS AND DISCUSSION

The dose at several locations around the ISS and the impact on the secondary particle contribution and energy is discussed. The efficiency of the VRT is presented, followed by recommended values for implementing the technique and steps for optimizing the simulation time depending on the user's objectives and dose uncertainty tolerance.

Geometry Dependence

To investigate the geometric dependence of the required isotropic region radius, the dose at each of the four detectors

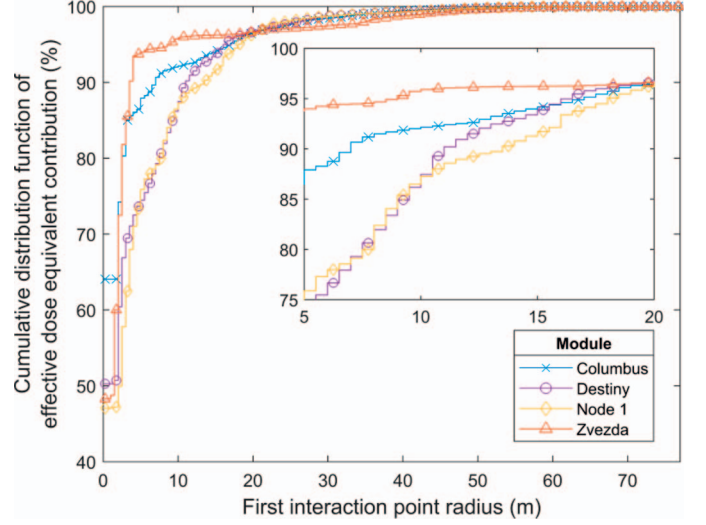


FIG. 4. Cumulative distribution function of effective dose equivalent by full GCR spectrum in the Columbus, Destiny, Node 1 and Zvezda modules. Isotropic regions up to 20m are displayed inset as it accounts for up to 97% of dose in each.

has been compared. The cumulative distribution function for dose as a function of the first interaction point radius, R_1 , was computed for all particles and summed. The results are presented in Fig. 4 and represent the upper bound on the error in the estimated dose from a simulation adopting a reduced value of R_{iso} corresponding to any value of R_1 , i.e., the error in the simulation will be less than or equal to the dose contribution from particles with $R_1 > R_{iso}$.

An isotropic radius of 50 meters will account for at least 99.5% of dose when compared to the analog simulation. Setting a radius > 50 meters will have a negligible additional contribution and is an inefficient use of computation time as there is a very low probability of a secondary particle interacting with the detector. A radius of 20 meters from the detector center is sufficient to retain at least 96.6% of dose, and its reciprocal (1.035) could be used as a correction factor to recover an estimate of the full analog model dose. If the isotropic region is too small, the contribution to dose is underestimated as many significant secondary particles will not be generated. Importantly, $R_{iso} < 20$ m would result in the doses at different locations being underestimated by substantially different fractions. An isotropic region radius of 5 meters or less is not recommended as the dose will be underestimated by up to 25.6%. If the isotropic region is the size of the detector (e.g., < 1 meter), the dose would be less than 50% of the full dose. This indicates that over 50% of dose contribution is from secondary particles.

Node 1 and Destiny have a larger dose contribution from secondary particles between 4 and 20 meters relative to the Columbus and Zvezda modules. The variation in dose contribution is attributable to the amount of passive shielding surrounding the location. Node 1 and Destiny both have larger amounts of material surrounding them that the GCR particles must be transported through, and therefore, a higher probability of interactions producing secondary particles. This includes adjacent habitable modules (Node 2, Leonardo, Zarya) and the

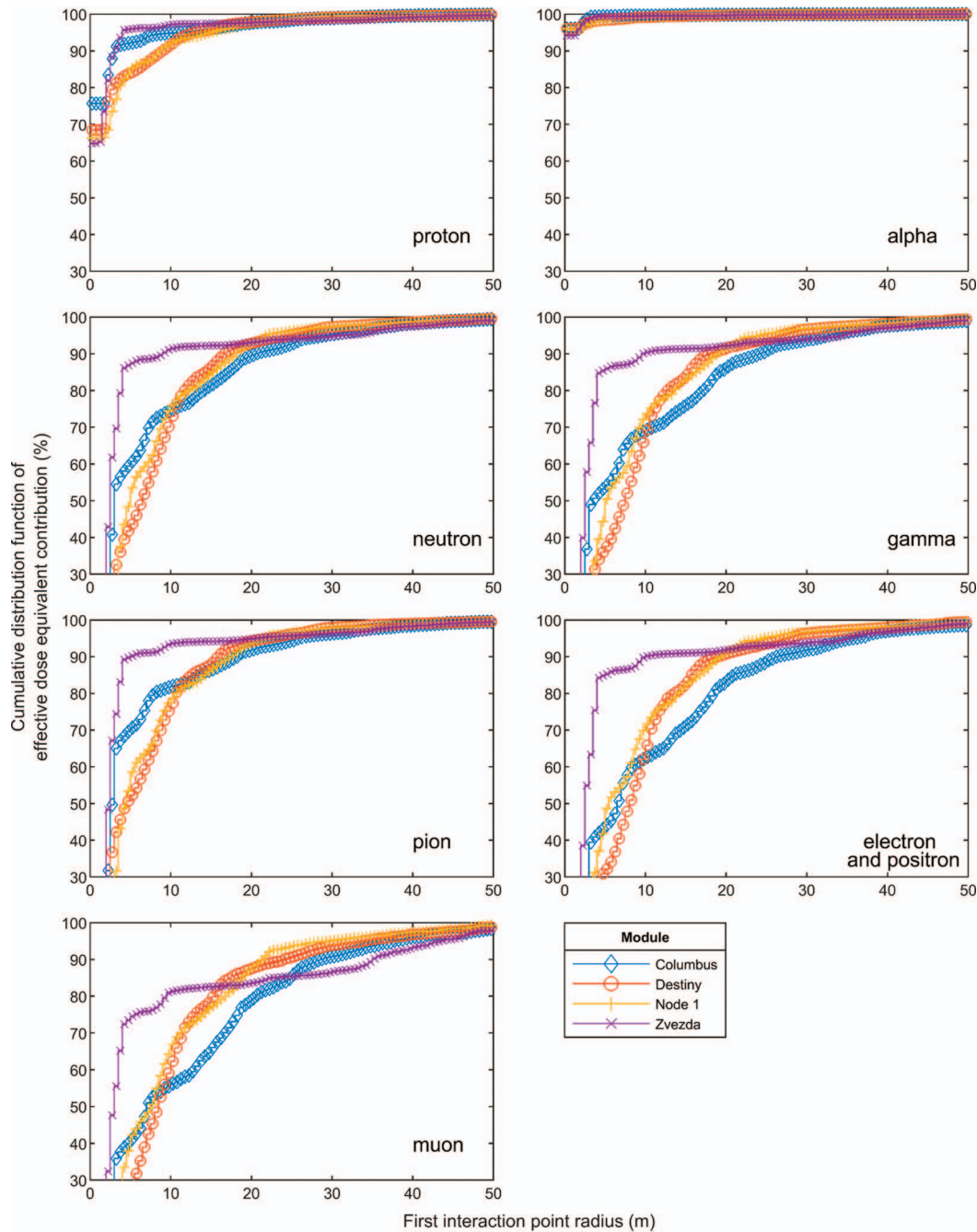


FIG. 5. Effective dose equivalent contribution by particle as a function of isotropic region radius (m) for Columbus, Destiny, Node 1 and Zvezda.

truss, whereas Columbus and Zvezda are located at the far forward and aft sections of the ISS. The Zvezda has the least shielding from adjacent modules. Therefore, Columbus and Zvezda can have a smaller isotropic region radius while retaining simulation accuracy.

Implementing the VRT will alter the secondary particle production, and consequently, their contribution to dose. Figure 5 can be used to determine the different contributions to dose depending on the secondary particle species. It represents the upper bound on error of the dose contribution by each particle type as a function of isotropic region radius.

Results are presented for Columbus, Destiny, Node 1 and Zvezda Service Module.

An isotropic radius of 5 meters is sufficient to retain at least 84.2% of dose contribution by protons (primary and secondary combined) and 98% of dose from alpha particles. The remaining secondary particles only account for less than 50% of their dose contribution, however, the combined contribution of gammas, muons, electrons and positrons is less than 5% at each dose point. Neutron contribution can vary between 14% and 23% depending on the module. To ensure at least 90% of neutron dose contribution, an isotropic radius of 20.75 meters is required.

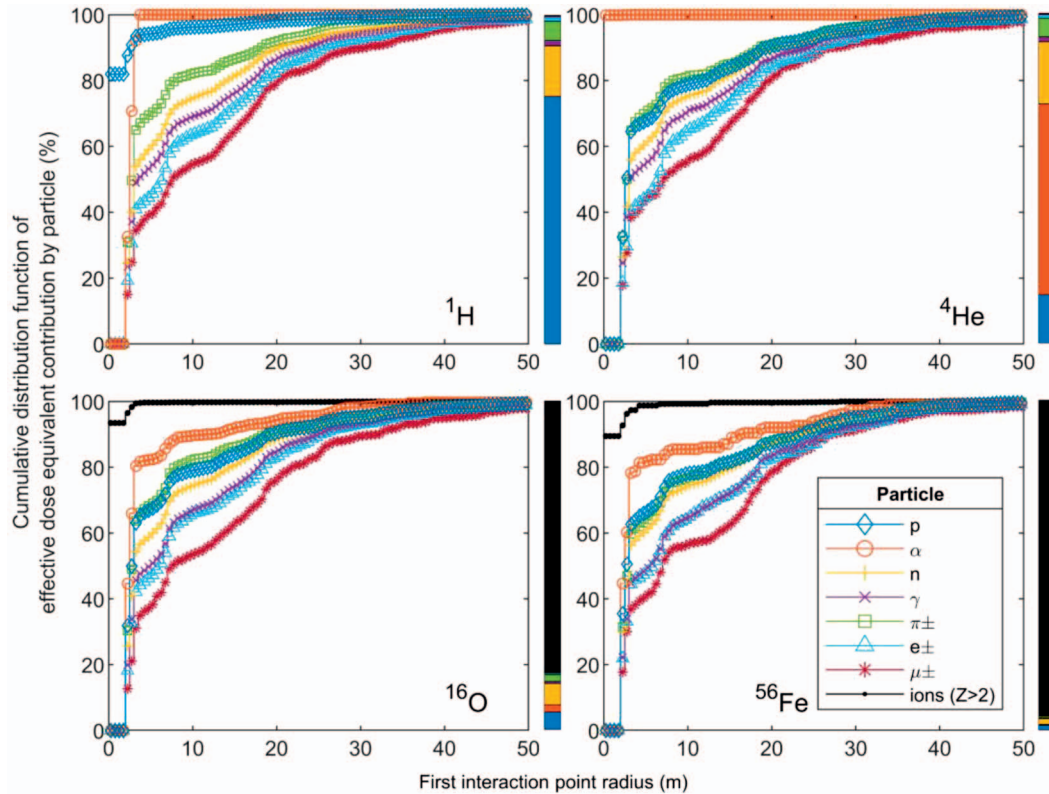


FIG. 6. Cumulative distribution function for effective dose equivalent contribution by particle as a function of isotropic region radius for four primary particles. Top-left panel: Proton, top-right panel: helium ions, bottom-left panel: oxygen ions, bottom-right panel: iron ions. The bar chart on the right of each plot shows the relative contribution to dose by each particle type. Note, the heavy ion contribution was not displayed in the proton and alpha plot as the dose contribution is negligible.

Particle Dependence

Figure 6 shows the cumulative distribution functions for several primary particle species (^1H , ^4He , ^{16}O , ^{56}Fe) and their secondary radiations. A similar dependence on R_1 can be seen for each secondary species and reflects their relative ranges in matter. The bar chart on the right of each plot shows the relative contribution to dose by each particle type. The figures can be used to inform the selection of isotropic radius for primary particles to ensure that sufficient secondary contribution is accounted for while obtaining a reduction in computation time. The region of isotropic irradiation by primary particles in the simulations can be specified individually for each primary particle type. The data shown in Fig. 6 demonstrate that there is opportunity for optimizing the choices of R_{iso} . This is important given that simulation times for heavy ions are up to 137 times longer than those for protons, so greater time reductions are achievable by selecting smaller R_{iso} for heavier ions. Note that no primary particle species should be restricted too far, as none is negligible. For example, for the Columbus module, 59% of total dose is from protons and their secondaries, 19% from alphas, and 23% from heavy ions. A similar breakdown is observed for other modules, within a few percent.

Low-LET secondary particles such as electrons, positrons, muons and gamma rays, traverse larger distances on average through spacecraft materials before reaching the

dose point. However, their contribution to dose is less than 10%. If simulating the low-LET particles is a priority then the user must ensure a larger isotropic region radius is used, such as 20 meters or greater.

From the Monte Carlo simulations of incident oxygen and iron ions, the dose contribution is above 99.5% even at low radii. This indicates that a majority of ion contribution will be simulated with a small region radius.

The angular distribution of fragments is dependent on primary particle energy, target material and target thickness, but they generally retain the direction of the incident nucleus. Less than 10% of dose is from secondary heavy ion fragments. The secondary contribution from remaining particles for incident oxygen and iron ions is 17.3% and 4.4%, respectively. As a result, reducing the isotropic region around the spacecraft will have negligible impact on heavy ion contribution to dose, but will underestimate their secondary particle production. A similar conclusion can be made for helium and oxygen ions, and assumed applicable for all heavier ions. For shorter computation time, an isotropic region radius of 10 meters is acceptable.

User Implementation

The following section discusses the steps required to implement the VRT to reduce computation time of Monte Carlo

TABLE 1

Recommended Input Parameters for VRT Implementation for Isotropic Protons, Alpha Particles, Ions ($3 \leq Z < 9$) and Heavier Ions ($9 \leq Z \leq 28$) for a Given Error Tolerance

Error tolerance (%)		1%	2%	3%	5%	10%
R_W (km)		10	10	10	10	10
Proton	R_{iso} (m)	44.75	36.75	31.25	19.25	15.25
	θ_{max} (°)	0.256	0.211	0.179	0.110	0.087
Alpha	R_{iso} (m)	43.75	35.75	26.25	18.75	12.75
	θ_{max} (°)	0.251	0.205	0.150	0.107	0.073
Ions ($3 \leq Z < 9$)	R_{iso} (m)	42.75	34.75	22.75	16.25	10.25
	θ_{max} (°)	0.245	0.199	0.130	0.093	0.059
Ions ($9 \leq Z \leq 28$)	R_{iso} (m)	22.75	15.75	11.75	9.25	3.75
	θ_{max} (°)	0.130	0.090	0.067	0.053	0.021

simulations. Recommended values for input parameters to implement the VRT are listed in Tables 1 and 2. The recommended values should only be used if the user's spacecraft geometry and materials resemble those of the ISS used in this analysis. If the objective of the simulations is to calculate total dose, use Table 1. Due to the scattering angle dependence on atomic number of the primary particle (as described in Particle Dependence section) a different isotropic region radius can be assigned for each primary particle species. Table 1 contains values for protons, alpha particles, ions ($3 \leq Z < 9$) and heavier ions ($9 \leq Z \leq 28$). If the user is interested in increasing resolution of low-LET particle contribution, Table 1 values are not recommended. For dose by particle, use Table 2. The error tolerances listed are the upper bound of the four locations simulated in the ISS to account for geometric and material variations in the spacecraft and are conservative for simulations of the ISS. The recommended minimum world radius (R_W) is 10 km and must be used for the maximum incident angle (θ_{max}). Supplementary Materials, Appendix 3 (<https://doi.org/10.1667/RADE-23-00027.1.S1>) lists the recommended commands for implementing the technique using Geant4's General Particle Source (GPS).

Depending on the user's objective of the simulations, the radius can be selected to obtain a variance reduction. For example, if the user is interested in designing passive shielding to reduce neutron dose, such as placing polyethylene slabs

(39), and requires a minimum of 95% of all neutrons simulated, then an isotropic region radius of 10 meters meets the specification. For active radiation shielding studies that require a majority of charged particles, such as protons, alpha particles and heavier ions, a 15-meter isotropic region radius will ensure over 95% of the dose contribution, but will not retain the low-LET particle contribution.

The recommended values are conservative and further optimization of input parameters can be achieved. The following steps can be used to determine input parameters.

1. Select an acceptable error tolerance for the dose (e.g., 2%, 5%, 10%). A higher error tolerance will result in higher efficiency and a larger time reduction as shown in Fig. 7.
2. Select an isotropic region radius (R_{iso}) for a given error tolerance using Figs. 5 and 6 or using data in the Supplementary Materials (<https://doi.org/10.1667/RADE-23-00027.1.S1>).
3. Calculate a world radius (R_W) using a minimum R_W/R_{iso} ratio of 100.
4. Calculate the maximum incident angle (θ_{max}) using Eq. (1).
5. Apply the input parameters and run the simulation.

For any choice of R_{iso} , Figs. 5 and 6 represent an upper bound on the consequential error. When a simulation is completed with a constrained isotropic radius, the error will be less than or equal to the values from the figures. To

TABLE 2

Total And Particle Values Represent Upper Bound Error in Effective Dose Equivalent Contribution for a Given Isotropic Region Radius (R_{iso})

R_{iso} (m)	5	10	15	20	25	30	35	40	45
θ_{max} (°)	0.029	0.057	0.086	0.115	0.143	0.172	0.201	0.229	0.258
R_W (km)	10	10	10	10	10	10	10	10	10
Total (%)	25.6	12.8	8.3	3.4	2.9	2.6	2.0	1.3	0.8
p (%)	15.8	7.8	4.9	2.4	2.0	1.8	1.4	0.9	0.6
α (%)	1.7	0.9	0.5	0.4	0.3	0.3	0.2	0.1	0.0
n (%)	56.4	27.0	18.3	9.9	7.4	5.3	4.0	2.6	0.9
γ (%)	62.4	30.8	23.7	13.5	9.0	6.0	4.6	3.0	1.9
$\pi \pm$ (%)	47.7	22.7	13.3	8.1	5.6	3.7	2.7	1.7	1.0
e $\pm$ (%)	67.9	37.4	28.4	16.3	11.1	8.1	4.9	3.2	2.0
$\mu \pm$ (%)	70.7	43.8	34.0	20.7	14.7	13.4	9.8	6.7	4.4

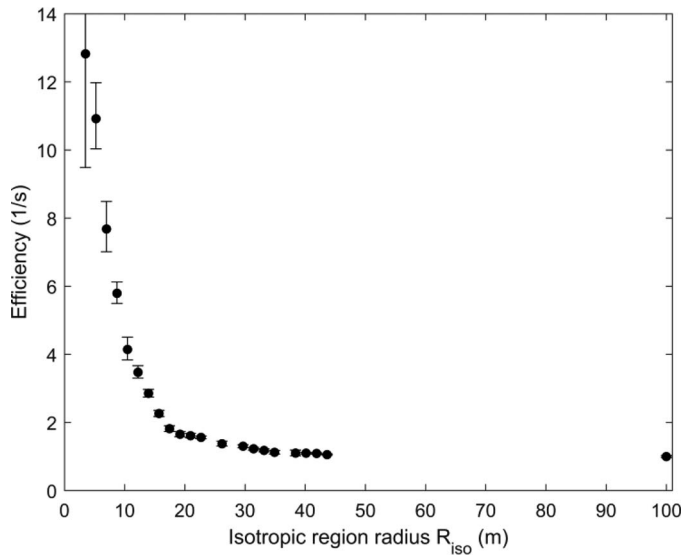


FIG. 7. Efficiency improvement ($1/T$) of the VRT as a function of reduced isotropic radiation field radius, R_{iso} , normalized to the full analog simulation ($R_{iso} = 100$ m).

verify this, several simulations were completed with a constrained isotropic radius. In all cases, the simulated dose with the VRT implemented was less than the estimated error bound. An isotropic region of 15 m around a dose point in the Columbus module, as predicted with the method described, has a maximum upper bound on the error of 6%. When the VRT was applied, the error was only 2.2%. Similarly, for a 10 m isotropic radius, the error was only 2.6%, far less than the 8% tolerance specified.

For users simulating spacecraft or habitat geometries and/or materials significantly different from those described in this work, one should conduct a similar analysis to determine appropriate isotropic region radii.

Variance Reduction and Efficiency

Computational efficiency in Monte Carlo simulations can be characterized by a metric such as that defined in Eq. (7). Under conditions in which the calculated quantity is not substantially different, σ^2 may be approximately constant and the efficiency improvement can be characterized by the inverse of calculation time, T , required to obtain a result comparable to the analogous full simulation. Figure 7 is a plot of normalized efficiency against the isotropic region radius.

A large improvement in simulation efficiency is evident with isotropic region radii less than 20 meters. Between 100- and 20-meter radius, an efficiency improvement of approximately 62% is observed and a 38% reduction in computation time. Computation time (and hence efficiency) is dependent on the geometry and material distribution around the detector, and therefore, varies for each detector location.

If modelling a non-isotropic context such as LEO or lunar orbit, directional planetary shadowing or geomagnetic shielding would not only change the isotropy but may impact the particle spectrum. The VRT processing time benefits may be different under these circumstances.

Computation time per particle history increases dramatically when particles are transported through material other than the vacuum of space. By reducing the isotropic region radii, the probability of a primary particle interacting with the materials of the spacecraft increases, and hence, total simulation time per history increases. As a result, the efficiency improvements achieved with the VRT increase at a much smaller rate than the improvement predicted by the solid angle reduction alone. Another factor contributing to timing is the efficiency of the user's software which for Geant4 includes the UserSteppingAction, UserStackingAction, UserTrackingAction and UserEventAction. In the custom Geant4 application developed, all fluence-to-dose conversion coefficients are applied during post processing of the data to minimize computation time.

CONCLUSION

Astronauts face severe health risks due to radiation during space flight and on board spacecraft. Multipurpose materials provide protection from all space risks, including radiation, but also generate secondary particles which contribute to overall dose. Improvements to spacecraft design for future space missions rely upon simulations of particle transport using the Monte Carlo secondary radiation dose from particles with a low LET. Heavy ions can benefit from a smaller isotropic region radius than protons and light ions as secondary particles generally retain the direction of the incident nucleus, and typically offer the greatest process time saving.

Recommended input parameters have been provided for isotropic protons, alpha particles, ions ($3 \leq Z \leq 9$) and heavier ions ($9 \leq Z \leq 28$) for a given error tolerance and can be easily implemented within Geant4 using the General Particle Source. Alternatively, the user can select an isotropic region radius from the figures presented.

Future work could involve completing a similar analysis for future spacecraft designs and also different outputs such as Total Ionizing Dose (TID) applicable to electronics. The method could also be applied to lunar surface or planetary surface Monte Carlo simulations to reduce computation time.

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